AI-Based Bank Loan Pre-Screening and Decision-Making System Using Rule-Based Expert Inference and Explainable Reasoning

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***Abstract*—Bank loan officers must decide quickly whether an application is worth sending to full underwriting, and they must be able to say why. Manual pre-screening is slow and inconsistent, and black-box credit models make the reasons hard to audit. This paper presents LoanSense AI, an explainable rule-based expert system that pre-screens Personal, Home, Car and Education loan applications. Applications are classified as Eligible, Ineligible or Incomplete using IF-THEN production rules, a completeness check, and a forward-chaining inference engine that evaluates every rule and records a plain-language reason for each pass or fail. The prototype is a Flask and SQLite web application with a Bootstrap interface, user authentication and a stored decision history. On a suite of 200 boundary, edge and missing-input test cases (50 per category) the engine agreed with the expected outcome in all 200 cases, with a mean latency of 11.9 ms per application. We discuss what this result does and does not show, because the expected outcomes are derived from the same rule specification, and we list the limitations of the current prototype.**

***Index Terms*—rule-based expert systems, loan pre-screening, explainable AI, decision-support systems, inference engine, financial knowledge base**

I. Introduction

Retail banks receive large numbers of credit applications every day. Before a human credit analyst spends time on in-depth underwriting, the bank runs a preliminary pre-screening that checks basic eligibility requirements [**?**]. Done by hand, this step is slow, depends on who happens to review the file, and handles missing or ambiguous inputs badly.

Planning a pre-screening tool combines three hard problems. First, each loan category has different criteria: Personal loans stress net monthly income and length of service; Home loans stress repayment capacity, credit score and the loan-to-value ratio of the property; Car loans stress income, employment stability and vehicle value; Education loans stress admission status, academic record and the co-borrower. Second, many applications arrive with fields missing, and a tool that guesses the missing values can approve or reject for the wrong reason. Third, an applicant or auditor who receives a "no" must be able to see which condition caused it.

The third demand is the reason this work avoids learned models. Rudin argues that for high-stakes decisions, interpretable models should be preferred to post-hoc explanations of black boxes [**?**], and explainable AI has been surveyed broadly [**?**], [**?**]. Credit decisions are such a setting, so we use classical symbolic AI in the sense of Russell and Norvig [**?**]: a knowledge base of production rules and a deterministic inference engine.

This paper presents LoanSense AI, a web application that follows the pre-screening chain from raw application data to a decision with an itemised justification. Its contributions are:

- **Multi-category knowledge base.** Structured IF-THEN rule sets for Personal, Home, Car and Education loans, built on three standard financial measures (FOIR, LTV and EMI), in which every decision traces to a stated rule.
- **Deterministic inference engine.** A forward-chaining engine that first checks input completeness, then evaluates every rule of the selected category, and returns one of three states: Eligible, Ineligible or Incomplete.
- **Explainable decision module.** A rule audit generator that stores a pass or fail reason for every rule, so the decision screen shows both what failed and what passed.
- **End-to-end micro-system and evaluation.** A Flask and SQLite application with authentication, input validation and history tracking, evaluated on 200 test cases for decision agreement, completeness detection and latency, together with an explicit discussion of the limits of that evaluation.

Section II reviews related work, Section III formulates the problem, Section IV describes the proposed system, Section V gives the implementation and experimental setup, Section VI reports the results and discussion, and Section VII concludes.

II. Related Work

A. Statistical and machine-learning credit scoring

Automated financial decisions have traditionally relied on operations research and statistical scoring models. Machine-learning methods have since shown strong predictive performance on consumer credit data [**?**], and gradient-boosted trees such as XGBoost [**?**] are widely used for tabular risk prediction. These models need large historical datasets, can degrade when economic conditions shift, and produce a score rather than a statement of which threshold an applicant missed. They are mostly accuracy-centred and do not by themselves produce a reason that a loan officer can read.

B. Expert systems and rule-based reasoning

Rule-based expert systems encode the knowledge of human specialists as production rules that an inference engine applies to the facts of a case. MYCIN-style systems are the classic example [**?**], and textbooks on expert systems cover the design of knowledge bases and inference engines [**?**], [**?**]. Forward chaining, which starts from known facts and fires rules until a conclusion is reached, is the standard strategy when the facts are given and the conclusion is wanted, as in eligibility screening; efficient matching algorithms such as Rete exist for large rule bases [**?**]. Rule systems are fully auditable by construction, but they are only as good as the rules written into them.

C. Explainability and regulation

Post-hoc methods such as SHAP [**?**] and LIME [**?**] approximate the contribution of each feature to the output of a complex model. Rudin cautions that such explanations can be inaccurate or misleading about what the model actually computes [**?**]. On the regulatory side, Article 22 of the GDPR restricts decisions based solely on automated processing [**?**], and its implications for explanation are discussed in [**?**]; whether it creates a full "right to explanation" is disputed [**?**]. In India, lenders are guided by the Reserve Bank of India's fair-practices requirements [**?**]. We therefore make a modest claim: an itemised, rule-level reason supports the kind of transparency these frameworks ask for, but we do not claim legal compliance.

D. Positioning

Table **??** positions this work. Among the works surveyed here, learned models offer accuracy but weak native explanations, and the classical expert-system literature offers transparency but is not applied to multi-category Indian retail loan pre-screening with an explicit Incomplete state.

Table I
Comparison with Related Work

Work	Focus	Technique	Learned model
[**?**]	Consumer credit risk	Machine-learning classifiers	Yes
[**?**]	Tabular prediction	Gradient-boosted trees	Yes
[**?**]	Medical consultation	Production rules	No
[**?**], [**?**]	Explaining model outputs	Post-hoc attribution	Applied to learned models
[**?**]	High-stakes decisions	Interpretable models preferred	Argues against black boxes
Ours	Loan pre-screening	Rules + forward chaining + reasons	No

III. Problem Formulation

A. Scope and decision states

The system pre-screens applications for four loan categories, $C = \{\text{Personal}, \text{Home}, \text{Car}, \text{Education}\}$. It does not approve credit: an Eligible result means only that the application passes the basic rules and may proceed to a human underwriter. An application is a tuple $A = \{x_1, x_2, \ldots, x_n\}$ of financial and demographic attributes. For each category c there is a required schema $S_c \subseteq A$. The output is one of three decision states: ELIGIBLE, INELIGIBLE or INCOMPLETE.

B. Completeness and rule evaluation

The completeness function is:

$$\Phi(A, c) = \begin{cases} 1 & \text{if every } x \in S_c \text{ is present and valid} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

If $\Phi = 0$, the system returns INCOMPLETE together with the set of missing attributes $M = \{x \in S_c : x \text{ is empty}\}$. If $\Phi = 1$, the application is passed to the rule base $R_c = \{r_1, \ldots, r_m\}$, where each rule maps A to $\{0, 1\}$. The decision is:

$$\Psi_c(A) = \begin{cases} \text{ELIGIBLE} & \text{if } \prod_{j=1}^{m} r_j(A) = 1 \\ \text{INELIGIBLE} & \text{otherwise} \end{cases} \tag{2}$$

The product of boolean rules is equivalent to a conjunction, so a single failed rule makes the application

Ineligible. The engine nevertheless evaluates all m rules, so that every failed condition is reported and not only the first.

C. Financial measures

Three measures are used inside the rules.

1) **Fixed Obligations to Income Ratio (FOIR):** Measures debt-service burden relative to income:

$$\text{FOIR} = \frac{\sum \text{existing EMIs} + \text{proposed loan EMI}}{\text{net monthly income}} \tag{3}$$

The threshold is FOIR ≤ 0.50, or 0.55 for high-income brackets ($> 1,00,000$/month).

2) **Loan-to-Value ratio (LTV):** Measures collateralisation for secured loans:

$$\text{LTV} = \left(\frac{\text{requested loan amount}}{\text{total asset market value}} \right) \times 100\% \tag{4}$$

The thresholds are LTV $\leq 80\%$ for Home loans and LTV $\leq 85\%$ for Car loans.

3) **Equated Monthly Installment (EMI):** Uses standard amortisation, with principal P, monthly interest rate r and tenure n in months:

$$\text{EMI} = \frac{P \cdot r \cdot (1+r)^n}{(1+r)^n - 1} \tag{5}$$

For a zero interest rate, the system uses EMI $= P/n$.

D. Evaluation measures

We report three measures: Decision agreement, Completeness detection, and Inference latency. We also check explanation coverage: every evaluated rule must contribute exactly one reason to the output.

IV. Proposed System

A. Architecture

LoanSense AI consists of four main layers:

- **Presentation Layer:** Web interface built with HTML5, CSS3, Bootstrap 5, and JavaScript for user interactions and real-time field validation.
- **Controller & Service Layer:** Flask web server managing routes, user sessions, and service invocation.
- **Rule Inference Engine:** Core engine executing completeness checks, production rule evaluation, and generating audit trails.
- **Data Persistence Layer:** SQLite database storing user profiles, loan applications, and complete historical audit traces.

B. Knowledge base

The knowledge base holds category-specific production rules summarized in Table **??**.

Table II
Production Rules by Loan Category

Category	Rule ID	Condition
Personal	R_P1	Age ≥ 21 and ≤ 60 years
Personal	R_P2	Net monthly income $\geq 25,000$
Personal	R_P3	Continuous employment ≥ 2 years
Personal	R_P4	CIBIL / credit score ≥ 650
Personal	R_P5	FOIR ≤ 0.50
Home	R_H1	Age ≥ 23 and ≤ 65 years at loan maturity
Home	R_H2	Combined family monthly income $\geq 50,000$
Home	R_H3	Property LTV $\leq 80\%$
Home	R_H4	CIBIL / credit score ≥ 700
Home	R_H5	Clear legal title and property approval status
Car	R_C1	Age ≥ 21 and ≤ 65 years
Car	R_C2	Net monthly income $\geq 20,000$
Car	R_C3	Vehicle LTV $\leq 85\%$
Car	R_C4	Employment stability ≥ 1 year in present org.
Education	R_E1	Confirmed admission in recognized institution
Education	R_E2	Co-borrower monthly income $\geq 30,000$
Education	R_E3	Co-borrower credit score ≥ 650
Education	R_E4	Academic record $\geq 60\%$

C. Inference engine

The engine executes a forward-chaining evaluation procedure detailed in Algorithm **??**.

Algorithm 1 Forward-chaining pre-screening

function EVALUATE(A, c)
  $M \leftarrow \{x \in S_c : x \text{ is empty or invalid}\}$
  if $M \neq \emptyset$ **then**
    return (INCOMPLETE, missing $= M$)
  end if
  $passed \leftarrow []$; $failed \leftarrow []$
  for each rule $r_j \in R_c$ **do**
    $(ok, reason) \leftarrow r_j(A)$
    if ok **then**
      append $reason$ **to** $passed$
    else
      append $reason$ **to** $failed$
    end if
  end for
  if $failed = \emptyset$ **then**
    return (ELIGIBLE, audit_trail $= passed$)
  else
    return (INELIGIBLE, reasons $= failed$, passed_checks $= passed$)
  end if
end function

Because the engine evaluates all m rules once, its cost is $O(m)$ per application, with $m \leq 5$ in the current knowledge base.

D. Explanation module

Every rule constructs an audit string based on the applicant's input parameters. For instance, an applicant for a Personal Loan evaluated with a calculated

FOIR of 0.51 will receive an itemized rejection reason: "FOIR 0.51 exceeds the 0.50 limit (R_P5)", alongside passed records for age, income, employment, and credit score.

E. Security and data integrity

Database operations utilize parameterized prepared statements to prevent SQL injection risks. Authentication credentials utilize SHA-256 password hashing alongside HTTP-only cookies for session protection.

V. Implementation and Experimental Setup

LoanSense AI is implemented in Python using Flask and SQLite, with a Bootstrap 5 front-end. The evaluation benchmark comprises 200 synthetic test cases across all four loan categories (50 per category), covering typical applications, boundary values, and missing input sets. Latency experiments were conducted on an Intel Core i7 workstation comparing the rule engine against XGBoost, Random Forest, and Multi-Layer Perceptron (MLP) baselines.

VI. Results and Discussion

A. Decision agreement

The inference engine achieved 100% decision agreement across all 200 test cases (Table **??**).

Table III
Benchmark Results by Loan Category

Category	Test cases	Decision agreement	Mean latency (ms)
Personal loan	50	100%	11.2
Home loan	50	100%	13.8
Car loan	50	100%	10.5
Education loan	50	100%	12.1

This result confirms implementation fidelity with respect to the underlying production rules specification.

B. Completeness detection

Every incomplete test case successfully returned an INCOMPLETE decision status identifying missing attributes without raising runtime exceptions.

C. Latency comparison

Table **??** summarizes performance comparisons with ML baselines.

Table IV
Inference Latency Benchmarks

Architecture	Memory footprint	Mean latency
XGBoost classifier	145 MB	42.5 ms
Random Forest (100 trees)	88 MB	28.1 ms
Deep neural network (MLP)	320 MB	65.0 ms
LoanSense AI (Rule Engine)	< 15 **MB**	**11.9 ms**

D. Limitations

Rule thresholds represent standardized placeholders rather than custom-tuned bank policies. Hard boundary constraints lack flexibility for borderline cases, which remains appropriate strictly for preliminary pre-screening stages.

VII. Conclusion and Future Work

LoanSense AI demonstrates an explainable, deterministic rule-based framework for pre-screening bank loan applications. Future extensions include dynamic threshold configuration portals, integration with document parsing pipelines, slow password hashing adoption, and formal user studies involving financial credit underwriters.

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